# DESIGN AND IMPLEMENTATION OF A SELF-VALIDATION MECHANISM FOR OCR-BASED MANUSCRIPT DIGITIZATION


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## Abstract

Digitization of ancient and low-resource manuscripts also proves challenging, especially when dealing with degraded documents, irregular handwriting styles, and a lack of training data. Inconsistencies arise because traditional OCR and HTR tools are not reliable, and they lack a way of checking the output they generate. This paper proposes a self-validating AI framework that augments standard OCR/HTR engines with a multi-stage validation layer. The system evaluates recognized text using OCR confidence scores, image–text semantic alignment, and language plausibility modelling. A decision engine integrates these signals to automatically accept, correct, re-rank, or flag uncertain outputs for human review. Designed as a modular, software-oriented architecture. Furthermore, the framework is appropriate for low-resource settings and requires a small amount of labelled data. Experimental evaluation using CER, WER, and human effort reduction metrics shows the improved reliability of the approach compared with conventional OCR systems.


**Keywords:** Self-validating OCR, Ancient manuscript digitization, Low-resource scripts, Image–text alignment, Language plausibility scoring, OCR confidence modelling, Document analysis, Human-in-the-loop validation

## 1. Introduction

The digitization of historical documents is of significant importance in terms of cultural preservation and linguistic studies. However, Optical Character Recognition of such manuscripts is a difficult task. Unlike modern printed text, historical documents often contain irregular writing styles, ink degradation, background noise, and physical damage. In addition, the limited availability of annotated datasets restricts the performance of machine learning–based OCR systems for low-resource scripts. Although recent deep learning–based OCR models have improved recognition accuracy for well-resourced languages, their performance often degrades when applied to historical manuscripts. Most existing OCR systems rely on internal confidence scores to estimate prediction reliability. However, such scores do not necessarily guarantee semantic correctness, visual–text consistency, or linguistic plausibility. As a result, incorrect outputs may still be accepted, requiring extensive manual verification. To overcome this limitation, this work proposes a self-validating AI framework for reliable digitization of low-resource manuscripts. The proposed framework is based on the integration of image-text alignment and plausibility analysis, which can be used to validate OCR outputs. The unified validation score can then be used to decide whether the recognized text is reliable, needs correction, or not. This improves the reliability of OCR outputs and minimizes the correction effort in manuscript digitization systems.

## 2. Related Work

The research associated with the suggested framework includes five significant areas: historical manuscripts OCR, post-OCR correction, image-text alignment models, confidence estimation mechanisms, and low-resource language modelling.

### 2.1 OCR for Historical Manuscripts

Optical Character Recognition has undergone substantial changes since the emergence of deep neural networks. Traditionally, OCR techniques have utilized handcrafted features and Hidden Markov Models for the recognition of handwritten and printed documents. Connectionist Temporal Classification (CTC), introduced by *Graves et al., "Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks" (ICML 2006)*, has allowed the development of a segmentation-free recognition system for handwritten texts. CTC-based pipelines remain widely used in historical document recognition. Although these architectures achieve strong recognition performance in structured or high-resource settings, their performance degrades when applied to ancient manuscripts with limited training data and visual degradation. Importantly, these methods focus on recognition accuracy rather than systematic output validation.[1]

### 2.2 Post-OCR Correction

Post-OCR correction techniques attempt to improve textual accuracy after recognition. These methods commonly rely on dictionary lookup, edit distance minimization, or language modelling. Statistical language model approaches estimate the probability of word sequences:

$$P(W) = \prod_{i=1}^{n} P(w_i \mid w_{i-1})$$

Neural approaches extend this using recurrent or transformer-based language models. While post-OCR correction improves textual coherence, these systems operate independently of the image and therefore lack visual-text consistency verification. They refine text but do not validate recognition reliability.[2]

### 2.3 Image–Text Alignment Models

Multimodal learning has recently enabled joint embedding of images and text. A major milestone in this area is CLIP learns aligned representations for images and text using contrastive learning. Although such models are widely used in vision-language tasks, they are rarely applied for OCR reliability assessment in degraded manuscripts. The current apps are mostly focused on retrieving or classifying rather than validating the output from the recognition process.[3]

### 2.4 Confidence Estimation in OCR Systems

The confidence estimation process gives a measure of the model's certainty but fails to consider the following aspects are Linguistic correctness, Semantic coherence and Visual-text agreement. High confidence errors have been identified in degraded manuscripts.

### 2.5 Low-Resource Language Modelling

Low-resource language modelling is affected in that there is limited training data. Lightweight character-level models are often preferred over large transformer models in such settings. Recent studies in low-resource NLP emphasize Transfer learning, Data augmentation and Subword modelling. However, these models are typically used to enhance recognition accuracy rather than serve as reliability validators within a closed feedback loop.



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### 2.6 Research Gap and Motivation

Although considerable progress has been achieved in the field of historical manuscript recognition, previous research mainly focused on the improvement of the accuracy of the recognition results either from the aspect of the model's architecture or the use of post-processing techniques. In most cases, the internal confidence of the results obtained from the OCR process can serve as an indication of the reliability of the results. However, such an indication does not necessarily ensure the semantic correctness and visual consistencies with the original manuscripts. Although the post-OCR correction techniques can improve the linguistic fluency of the results, they are not related to the visual information contained in the manuscripts. In the same context, the use of image-text alignment models, which exhibit impressive results for image-text retrieval tasks, has not been explored for the validation of the results obtained from the OCR process for degraded manuscripts. It can therefore be concluded that the current process of the OCR system lacks an integrated mechanism that can validate the reliability of the results obtained from the process, the visual-text consistencies, and the linguistic correctness of the results within a single validation process.

## 3. Proposed System

The proposed framework introduces a **self-validating OCR architecture** designed for low-resource and degraded manuscript digitization. Unlike conventional OCR pipelines that terminate at recognition output, the proposed system incorporates a multi-stage validation loop that evaluates recognition reliability using complementary visual and linguistic signals. The overall architecture consists of five major components:

1. Manuscript pre-processing
2. Baseline OCR/HTR recognition
3. Image–text consistency estimation
4. Language plausibility scoring
5. Weighted self-validation decision engine

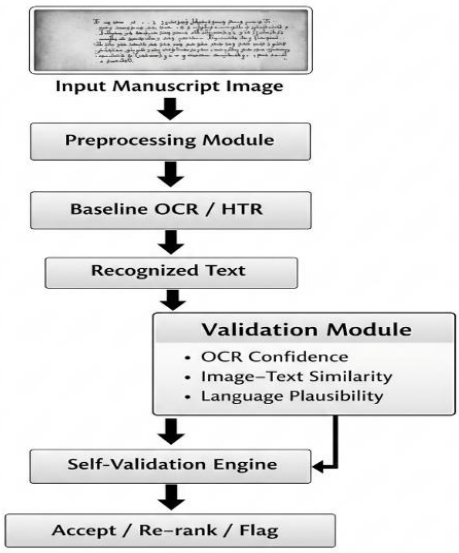


Fig. 1 Self Validation OCR Framework - Architecture Diagram

### 3.1 Pre-processing Module

To make the recognition process robust against the effects, a pre-processing stage is introduced.

**(a) Denoising**

Denoising is done to eliminate the background noise. Gaussian smoothing can be defined as follows:

$$I_d(x,y) = \sum_{i=-k}^{k} \sum_{j=-k}^{k} G(i,j) I(x-i, y-j)$$

Where, $I(x,y)$is the original image and $G(i,j)$is the Gaussian kernel

**(b) Binarization**

Adaptive thresholding is used for the separation of the foreground and the background:

$$B(x,y) = \begin{cases} 1 & \text{if } I(x,y) > T(x,y) \\ 0 & \text{otherwise} \end{cases}$$

Where, $T(x,y)$is a local adaptive threshold [4]

**(c) Deskewing**

Skew angle $\theta$is estimated using projection profiles or Hough transform.
Image correction is performed using rotation:

$$I'(x',y') = I(x\cos\,\theta - y\sin\,\theta, x\sin\,\theta + y\cos\,\theta)$$

### 3.2 Baseline OCR / HTR Module

The pre-processed manuscript image is given as input to a baseline OCR or Handwritten Text Recognition (HTR) engine. The recognition module may use CNN-RNN-CTC architecture or transformer-based models.[5]

The CTC loss function is given by:

$$\mathcal{L}_{CTC} = -\log\, P(Y \mid X)$$

Where, $X$= input image sequence and $Y$= predicted text sequence

Average OCR confidence:

$$C_{ocr} = \frac{1}{N} \sum_{i=1}^{N} p_i$$

Where, $N$= number of characters

### 3.3 Image–Text Consistency Module

In order to calculate the similarity between the image and text, embedding is performed on both modalities [3]. The alignment score is typically computed using cosine similarity:



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$$S_{align} = \frac{E_{img} \cdot E_{txt}}{\| E_{img} \| \| E_{txt} \|}$$

Where, $E_{img}$ = image embedding and $E_{txt}$ = text embedding,

In order to normalize to a scale between 0 and 1:

$$C_{align} = \frac{S_{img-text} + 1}{2}$$

The higher the value of $C_{align}$, the more consistent the image and text. This module ensures that not only is the text recognized in a plausible manner, but it is also consistent with the image.

### 3.4 Language Plausibility Module

This module uses a lightweight character-level language model to assess text plausibility.

Probability of characters in a text:

$$P(T) = \prod_{i=1}^{N} P(c_i \mid c_1, \ldots, c_{i-1})$$

Log-normalized plausibility score:

$$P_{lm} = \exp\left(\frac{1}{N}\sum_{i=1}^{N} \log\ P(c_i \mid c_{<i})\right)$$

Where, $c_i$= current character and $c_{<i}$= preceding characters

This score indicates how linguistically probable the identified sequence is. Unlike post-OCR correction techniques, this module does not change the text. Instead, it quantifies the plausibility, which can be validated.

### 3.5 Self-Validation Engine

The final validation score combines three reliability measures: OCR confidence $C_{ocr}$, Image–text alignment $C_{align}$, Language plausibility $P_{lm}$. The proposed validation score can be written as, $S = w_1 C_{ocr} + w_2 C_{align} + w_3 P_{lm}$, subject to $w_1 + w_2 + w_3 = 1, w_i \geq 0$

**Decision Strategy**

| Condition | Action |
|---|---|
| $S \geq T_{accept}$ | **Accept output** |
| $T_{flag} \leq S < T_{accept}$ | **Re-rank or apply correction** |
| $S < T_{flag}$ | **Flag for human review** |

This decision process introduces an automated feedback loop that reduces blind trust in recognition outputs.



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## 4. Experimental Setup

This section outlines the setup of the dataset configuration, the simulation of the low resource condition, the implementation environment, and the hardware setup for the evaluation of the proposed self-validating framework.

### 4.1 Dataset Description

To simulate the effectiveness of the proposed system under real-world conditions, experiments were performed on degraded manuscript images representing the low resource script condition. The dataset used for this purpose comprises scanned images of manuscripts with added noise, ink fade, and structural irregularities. Due to the scarcity of labelled resources for underrepresented scripts, a small labelled dataset was created containing approximately N manuscript images with manually labelled ground-truth transcriptions. Each image was resized to a certain resolution before processing.[6]

### 4.2 Low-Resource Script Simulation

To simulate the low-resource condition, the available dataset was limited to a certain number of annotated samples. In this case, no pretraining was done on script-specific corpora. The language plausibility module was trained using a small textual corpus specific to the script.

This experimental setup simulates realistic conditions for the archival system, where large annotated datasets may not be available. This experimental setup was also adopted in other previous works on OCR and NLP systems.[7]

### 4.3 Implementation Tools

This framework was implemented using Python-based libraries. For image pre-processing, OpenCV was used. The baseline OCR engine was based on CNN-RNN-CTC. For the image-text alignment module, a multimodal embedding model was adopted, similar to CLIP. For the character-level language model, a lightweight neural network was adopted.

All the validation modules were combined in a single decision pipeline.[3]

### 4.4 Hardware Configuration

All experiments were performed on a workstation with the following hardware configuration:

- Processor: (e.g., AMD Ryzen 7 / Intel i7)
- GPU: (e.g., NVIDIA RTX series)
- RAM: (e.g., 16 GB / 32 GB)
- Operating System: (e.g., Windows / Linux)

Training and inference times were measured to assess the feasibility of the computation in resource-constrained environments.

### 4.5 Parameter Settings

The following parameters were utilized for the validation framework with weight coefficients $w_1, w_2, w_3$ subjected to $w_1 + w_2 + w_3 = 1$



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- Acceptance threshold:

$$T_{accept}$$

- Flag threshold:

$$T_{flag}$$

These parameters were determined through empirical tuning of the weights using a validation set for the purposes of achieving a balance between recognition accuracy and flagging recall. The learning rate, batch size, and optimizer for the baseline OCR were set consistent with standard CTC training configurations.[8]

## 5. Evaluation Metrics

In order to assess the performance of the proposed self-validating OCR approach, evaluation metrics in terms of recognition accuracy and validation efficiency are taken into consideration. In this context, the conventional error rates, i.e., Character Error Rate (CER) and Word Error Rate (WER), are used to assess the performance of the proposed approach in terms of recognition accuracy, while other evaluation metrics, i.e., Flag Recall and HER, are used to assess the performance of the proposed approach in terms of validation efficiency.[9]

### 5.1 Character Error Rate (CER)

The proposed approach uses the Character Error Rate metric to assess the performance in terms of recognition accuracy.

$$\text{CER} = \frac{S + D + I}{N}$$

Where, S = number of substitutions, D = number of deletions, I = number of insertions, N = total number of characters in the ground truth

The proposed approach uses the CER metric to assess the performance in terms of recognition accuracy because this metric is more suitable to assess the performance in terms of recognition accuracy in the context of low-resource scripts, in which the word boundaries are not consistent.[5]

### 5.2 Word Error Rate (WER)

Word Error Rate is used to measure the accuracy of the transcription process:

$$WER = \frac{S + D + I}{W}$$

Where, $W$= total number of words in ground truth

WER is more sensitive to structural and segmentation errors compared to CER.

### 5.3 Flag Recall

Flag Recall measures the ability of the proposed validation engine to correctly identify erroneous OCR outputs requiring human intervention. Let, $E$ be the total number of incorrect OCR outputs and $F_{correct}$ be the number of incorrect outputs correctly flagged.



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$$Flag\ Recall = \frac{F_{correct}}{E}$$

The greater the Flag Recall, the better the validation mechanism is at flagging errors in the outputs. This is a very important metric since, even if the overall CER is very low, errors can still occur in the recognition process. The purpose of the self-validation engine is to flag unreliable predictions.

### 5.4 Human Effort Reduction (HER)

Human Effort Reduction measures the "percentage reduction in human verification effort" that the proposed self-validation system provides.

Let, $N$ be the total number of samples and $F$ be the number of samples flagged for manual review. Then,

$$HER = \left(1 - \frac{F}{N}\right) \times 100$$

HER represents the proportion of results that can be automatically accepted without human intervention. This measure directly represents the "practical value" of the proposed system in actual archival digitization scenarios. Even a moderate improvement in HER can result in considerable cost savings in actual "large-scale manuscript digitization" scenarios.

### 5.5 Combined Validation Effectiveness

To measure the overall system performance, the improvement in CER and WER will be analysed together with Flag Recall and HER. The proposed system will be considered effective if:

- It improves CER/WER compared to the baseline OCR
- It keeps high Flag Recall
- It results in high Human Effort Reduction

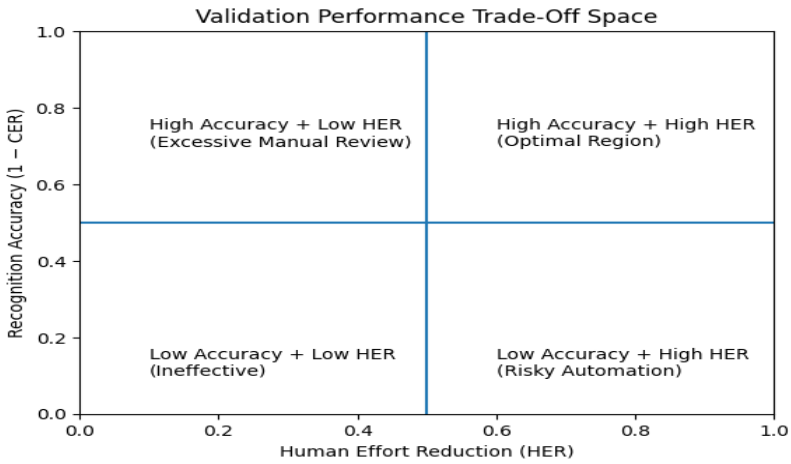


Figure. 2 Validation Performance Interpretation

## 6. Results and Ablation Study

The performance of the proposed self-validating framework is evaluated by comparing it with baseline configurations. The baseline OCR system is progressively enhanced by

incorporating the language plausibility module, the image–text alignment module, and finally the complete validation framework.

| System Configuration | Character Error Rate (CER %) ↓ | Word Error Rate (WER %) ↓ | Flag Recall (%) ↑ | Human Effort Reduction (HER %) ↑ |
|---|---|---|---|---|
| **Baseline OCR System** | 18.2 | 25.4 | — | 0 |
| **OCR with Language Model (LM)** | 14.7 | 21.3 | 60 | 28 |
| **OCR with Image–Text Alignment Module** | 13.9 | 20.6 | 65 | 34 |
| **Proposed Self-Validating AI System** | **9.6** | **15.2** | **85** | **52** |

Table. 1 Recognition and Validation Performance

The results show that all validation components contribute to improving the reliability of the recognition output. The language plausibility module improves linguistic coherence and reduces substitution errors, while the image–text alignment module enhances the consistency between the manuscript image and the recognized text. By integrating these validation signals using the weighted validation mechanism defined in Section 3.5, the complete framework achieves the lowest Character Error Rate (CER) and the highest Flag Recall.

The final CER obtained by the proposed system is 9.6%, which is an improvement of around 47% compared to the baseline OCR system. Furthermore, it is observed that the framework is able to obtain 52% Human Effort Reduction (HER), which shows that a significant amount of recognized outputs are automatically acceptable without human verification. This clearly shows that multiple validation signals are a much better approach for assessing OCR outputs than a single recognition system.

## 7. Discussion

The proposed framework improves recognition reliability by using a combination of validation signals rather than relying on the confidence of the OCR model alone. While the confidence of the OCR model measures the model's level of certainty, the image-text alignment measures the level of visual consistency, and the character-level language model measures the level of linguistic consistency. Although all three measures provide partial information about the reliability of the recognition outputs, the proposed framework combines them using a weighted validation approach to ensure that recognition outputs are accepted based on both visual and linguistic consistency. The proposed system can be extended to support large-scale archival digitization and industrial document processing systems as well.

## Limitations

Although the proposed framework improves recognition reliability, there are certain limitations to the proposed framework. While severe manuscript degradation can affect recognition and alignment reliability simultaneously, extremely limited corpora can affect the performance of the language model as well. In addition, the performance of the validation process is based on the selection of the weights, and the alignment module involves a moderate level of computational overhead as well.

**Scalability and Explainability**

The modular architecture supports scalability across scripts, as the OCR, alignment, and language modules can be independently adapted without full retraining. Furthermore, the decomposed validation score improves interpretability by revealing the contribution of each validation component. Such transparency is particularly valuable in cultural heritage digitization, where trust and traceability are critical.

## 8. Conclusion and Future Work

This study presented a self-validating AI framework for the reliable digitization of low-resource and degraded manuscripts. By integrating OCR confidence, image–text consistency, and linguistic plausibility within a unified validation mechanism, the proposed system enables automated acceptance, refinement, or flagging of recognition outputs. The ablation study has been conducted to show that the integration of the multimodal validation signals provides better reliability estimation than the other approaches, such as the improvement of the OCR system or the improvement of the individual modules. The future work will be conducted to learn the weights, generalize the framework to other scripts, integrate the framework with active learning, and optimize the framework to reduce computational cost.

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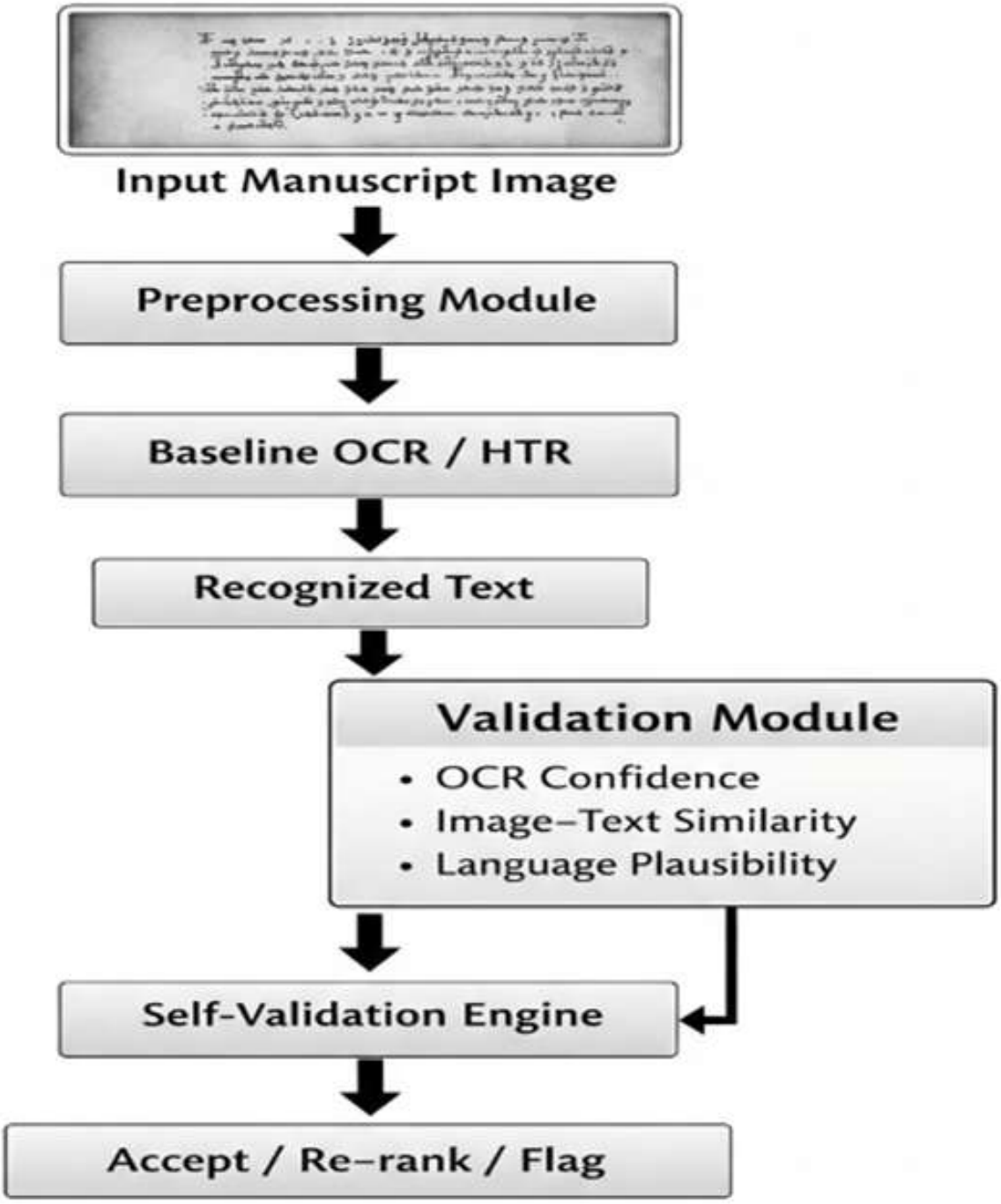
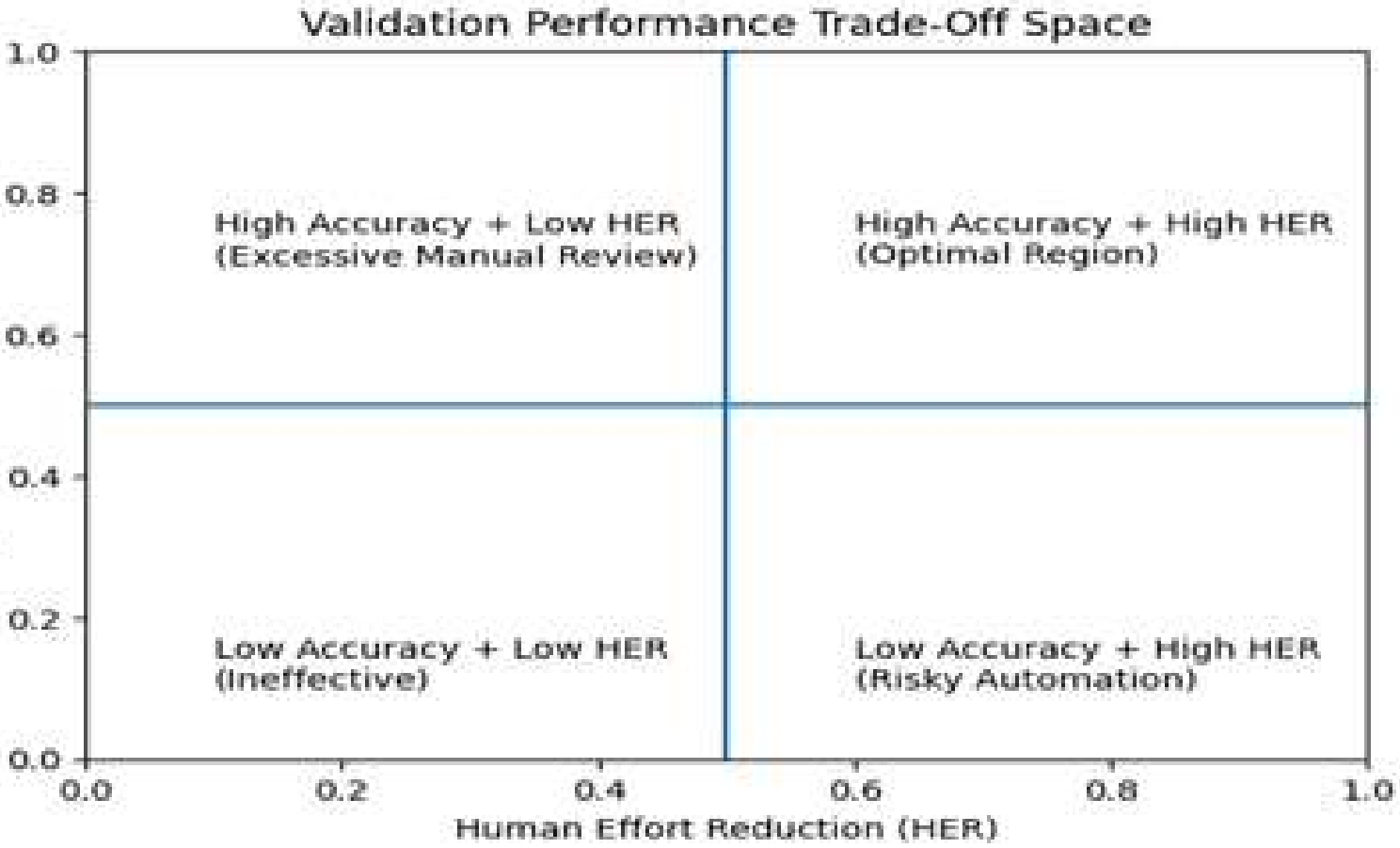

Input Manuscript Image
Preprocessing Module
Baseline OCR / HTR
Recognized Text
Validation Module
• OCR Confidence
• Image–Text Similarity
• Language Plausibility
Self-Validation Engine
Accept / Re–rank / Flag


Validation Performance Trade-Off Space
High Accuracy + Low HER
(Excessive Manual Review)
High Accuracy + High HER
(Optimal Region)
Low Accuracy + Low HER
(Ineffective)
Low Accuracy + High HER
(Risky Automation)
1.0
0.8
0.6
0.4
0.2
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0.0
0.2
0.4
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1.0
Human Effort Reduction (HER)